



Inside the Earth — Four Layers

S6E5a

Layer	What It's Like	State
Crust	Thinnest layer — the rocky outer shell where we live (like an apple's skin)	Solid
Mantle	Thickest layer — hot rock that flows VERY slowly	Solid but slowly flowing
Outer Core	Iron & nickel metal, hot enough to be liquid	Liquid
Inner Core	Iron & nickel; hottest part — squeezed SOLID by huge pressure	Solid

Going deeper, three things go UP: temperature (hotter), pressure (more squeezing), density (heavier). The inner core is nearly as hot as the Sun's surface!



ACTIVITY 1: Layer Logic

1. Thinnest layer, where we live: _____
2. Thickest layer, slow-flowing rock: _____
3. The only LIQUID layer: _____
4. Hottest layer, but solid: _____
5. Why is the inner core solid? _____
6. Going deeper, temperature goes: up / down (circle)



Minerals vs. Rocks

S6E5b

MINERAL

A naturally formed solid with ONE specific chemical makeup — a single pure **ingredient**. (Quartz, gold, salt)

ROCK

A solid made of one or MORE minerals mixed together — the finished **recipe**. (Granite = quartz + feldspar + more)

Mineral ID Test	What It Tells You
Hardness	How easily it scratches (diamond = hardest mineral)
Color & Streak	Its color, and the color of its powder when scraped
Luster	How shiny it is (metallic, glassy, dull)
Cleavage	How it breaks (smooth flat faces, or rough edges)



The Rock Cycle — Three Rock Types

S6E5c

Rock Type	How It Forms	Examples
Igneous	Melted rock (magma below ground / lava above) cools & hardens	Granite, obsidian, basalt
Sedimentary	Bits of rock & shell settle in layers and get pressed together	Sandstone, limestone
Metamorphic	Existing rock changed by heat & pressure (without melting)	Marble, slate

Decode the names: Igneous = "fire" (heat-made). Sedimentary = "sediment" (settling bits). Metamorphic = "changed form" (like metamorphosis — caterpillar → butterfly!). Any rock can become any other rock over time.



ACTIVITY 2: Rock Type Detective

Write **I** (igneous), **S** (sedimentary), or **M** (metamorphic):

1. Formed when lava cooled and hardened
2. Made of layers pressed together over time
3. Limestone changed by heat & pressure into marble
4. Sandstone — settled sand squeezed into rock
5. Obsidian — cooled so fast it looks like glass
6. The type where most FOSSILS are found

Flip over for weathering & erosion, plate tectonics, fossils & soil! →

Weathering → Erosion → Deposition S6E5d,e

1. WEATHERING breaks it → 2. EROSION moves it → 3. DEPOSITION drops it

Step	What Happens	Examples
Weathering	Rock breaks into smaller pieces right where it sits	Ice cracking rock; roots splitting stone; acid rain
Erosion	Pieces are carried away by water, wind, ice (glaciers), or gravity	Rivers & waves moving sediment; landslides
Deposition	Pieces are dropped in a new place	Sand building a beach; silt forming a delta

 **Don't confuse:** Weathering = **wear down in place** (no moving!). Erosion = **exit** — it moves away. Moving water is the most powerful agent — it carved the Grand Canyon!  **Humans change the surface too:** mining, building, farming, clearing forests (removing plants speeds up erosion; terracing slows it).

ACTIVITY 3: W, E, or D?

Write **W** (weathering), **E** (erosion), or **D** (deposition):

- | | |
|---|--|
| _____ 1. Ice freezes in a crack and splits the rock | _____ 4. Wind blows soil off a dry field |
| _____ 2. A river carries sand downstream | _____ 5. Tree roots break a sidewalk apart |
| _____ 3. Sand settles and builds up a delta | _____ 6. Waves drop sand to build a beach |

Plate Tectonics S6E5f

Earth's crust is cracked into giant **tectonic plates** that float on the slowly-flowing **mantle**. They move a few centimeters a year — fingernail speed — but over millions of years they reshape the planet.

Plate Movement	What Happens	Result
Push together (collide)	Crust crumples up, or one plate dives under another	Mountains & volcanoes
Pull apart (separate)	Magma rises to fill the gap → new crust	Mid-ocean ridges, rift valleys
Slide past	Plates grind, catch, then suddenly slip	Earthquakes

 **Earthquake** = sudden shaking when plates slip & release energy. **Volcano** = opening where **magma** reaches the surface (then it's called **lava**). Both cluster at plate **EDGES** — the "**Ring of Fire**" circles the Pacific!

Fossils & Soil S6E5g,h

Fossils = proof of change

Preserved remains of ancient life, usually in **sedimentary rock**. Sea fossils on mountaintops → that land was once underwater, pushed up by plates! **Deeper layers = older fossils**.

Soil = rock + life

Weathered rock + decayed plants/animals (**organic matter / humus**). Profile top → bottom: **topsoil** (dark, rich, roots) → **subsoil** (clay & minerals) → **weathered rock** → **bedrock** (solid).

ACTIVITY 4: Plates, Fossils & Soil Quiz

- | | |
|--|--|
| 1. Plates float on the slow-flowing: _____ | 4. Melted rock at the surface is called: _____ |
| 2. Plates colliding build: _____ | 5. Sea fossils on mountains prove: _____ |
| 3. Plates sliding past cause: _____ | 6. The dark, rich top layer of soil: _____ |

I Can... (Self-Check)

- | | |
|--|---|
| ☐ Name Earth's 4 layers and their states | ☐ Put weathering → erosion → deposition in order |
| ☐ Explain what increases with depth (heat, pressure, density) | ☐ Match plate movements to mountains, ridges & quakes |
| ☐ Tell a mineral (ingredient) from a rock (mixture) | ☐ Use fossils as evidence that Earth's surface changed |
| ☐ Explain how all 3 rock types form | ☐ Describe what soil is made of and its layers |